

For $Y_1, Y_2, \dots, Y_n$, there are totally n variance terms
and $n^2 - n$ covariance terms for:
 $\text{Var}(\sum_{i=1}^n Y_i)$.

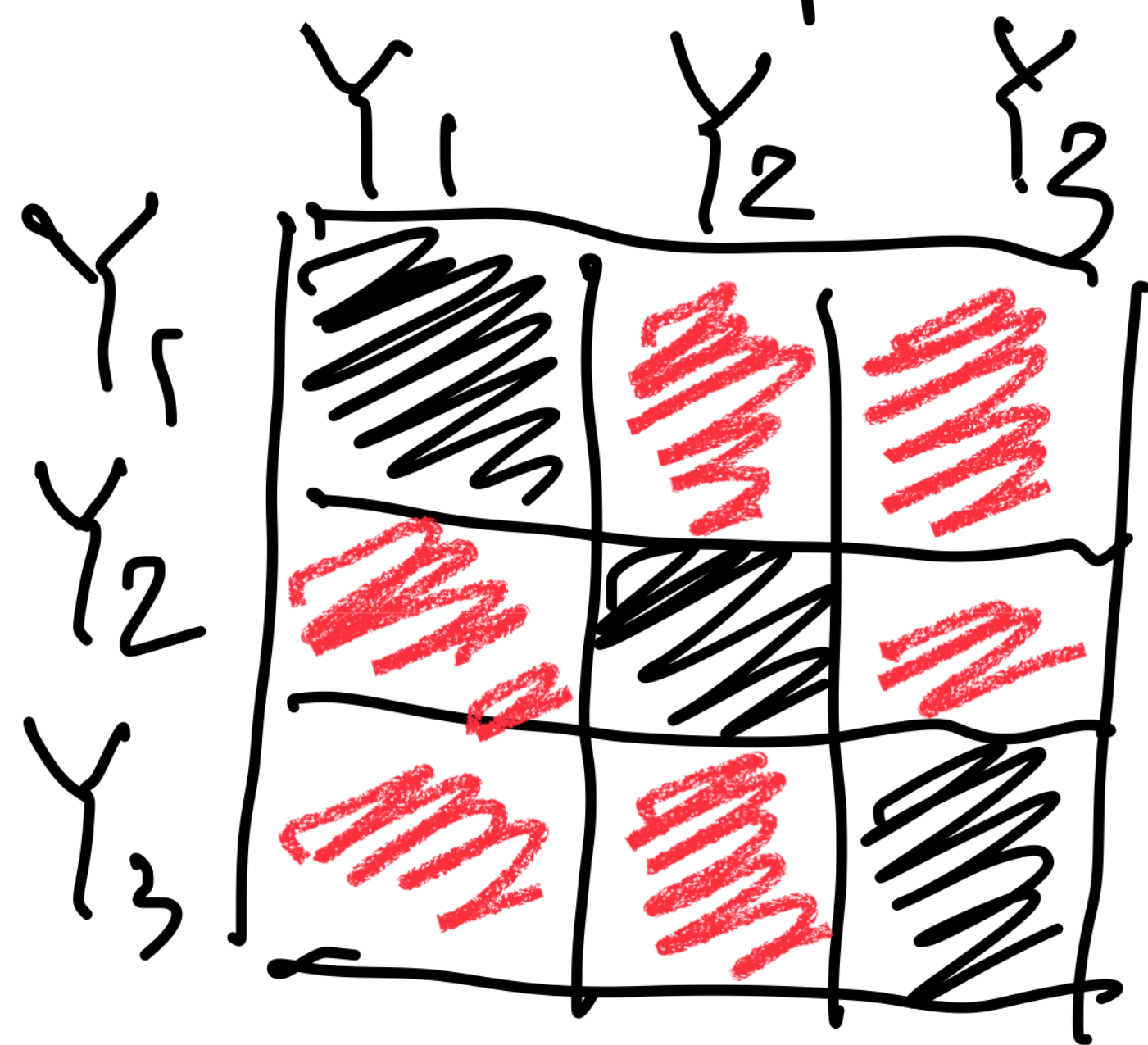
E.g., when $n=2$,

$$\text{Var}(Y_1 + Y_2) = \text{Var}(Y_1) + \text{Var}(Y_2) \\ + \text{Cov}(Y_1, Y_2) + \text{Cov}(Y_2, Y_1).$$

When $n=3$,

$$\text{Var}(Y_1 + Y_2 + Y_3) = \text{Var}(Y_1) + \text{Var}(Y_2) + \text{Var}(Y_3) \\ + \text{Cov}(Y_1, Y_2) + \text{Cov}(Y_1, Y_3) + \text{Cov}(Y_2, Y_3) \\ + \text{Cov}(Y_2, Y_1) + \text{Cov}(Y_3, Y_1) + \text{Cov}(Y_3, Y_2).$$

This corresponds



→ the following diagram:

i.e., diagonal terms $\Rightarrow$ variances
off-diagonal terms $\Rightarrow$ covariances.

$\Rightarrow$ In general, we have $n^2 - n$ covariance terms.